Game Development Technologies Research Workbook

Game Engines

1. Identify 3 different game engines that can be used to create 3D video games.
   Research each game engine and provide a list of features (at least 5) for each engine.
   One of these engines must be the Unreal Engine 4.

Game Engine	Features
Unreal Engine 4	Blueprint Visual Scripting: This allows for designers to design entire games without writing code, allowing for easy prototyping.
	Physically Based Rendering: Allows for high-fidelity graphics that simulates how light works in the real world.
	Real-Time Global Illumination and Lighting: Implements dynamic lighting allowing for more detailed shadows and reflections in real time.
	Full Source Code Access: Provides access to C++ source code, allowing users to customise and debug the engine to their needs.
	Platform Versatility: Supports deployment on Windows, Mac, Linux, PlayStation 4/5, Xbox One/Series X, iOS, Android, and Nintendo Switch.
Unity	2D & 3D Development: Supports both 2D and 3D game development through dedicated tools
	Visual Scripting & Tools: Inbuilt Shader Graph and Visual Effect Graph for easy creation of shaders and effects without using code.
	Asset Store: Premade models, audio, and texture to assist in development.
	Unity AI: Inbuilt AI for code generation and asset creation.
	Multiplayer Services: Allows for the creation and scaling of networked games.
Godot	Dedicated 2D and 3D Engines: Features 2 separate engines that are designed for either 2D or 3D games.
	Open Source: The engine is completely free to use!
	Deployment: Easy exportation to desktop, mobile and web.
	Node-Based Scene System:

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	Godot organises game logic into scenes, as a collection of nodes that can be manipulated to build game structures.
	GD Script and Language support: Features support for the use of GD Script as well as C#.

2. What makes each engine different?
   Some of the features you identified may be common across two or more engines. For each game engine identified, write a short statement about what makes this engine unique and why a game studio may want to use it. (Write one or two sentences for each engine)

Game Engine	Unique Selling Point
Unreal Engine 4	Unreal Engine 4 is defined by its high-end performance and capabilities, making it an industry standard for AAA games and titles. Its visual scripting and advanced rendering are especially useful for larger projects with a large budget.
Unity	Unity is extremely versatile as a dominant engine for mobile, VR and AR development. It primarily uses C# as well as having a massive Asset store with over 70,000 packages.
Godot	Godot is free to use and open source under the license of MIT allowing for developers to completely own their code for free. Godot is around 100MB and runs instantly without installation.

3. For each game engine identified, list sources of information for developers to refer to. This can be online technical information provided by the developer, how-to articles and blogs, YouTube channels, or textbooks.
   Identify at least 2 different sources for each engine.

Game Engine	Sources of Information
Unreal Engine 4	https://dev.epicgames.com/documentation/en-us/unreal-engine/find-help-and-answers?application_version=4.27
	Unreal Engine Tutorials - YouTube
Unity	https://support.unity.com/hc/en-us
	https://discussions.unity.com/
Godot	https://godotengine.org/blog/
	https://godotengine.org/community/

Image Editing Software

4. Identify 3 different programs used for image editing.
   These may be PC applications, web services, or mobile apps.
   Research each program and provide a list of features (at least 3) for each.
   One of these programs must be Krita.

Image Editing Software	Features
Krita	**Advanced Brush Engines:** Includes over 100 preloaded brushes that can be user customised to create desired effects.

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	File Format Support: Krita supports a vast range of formats including PSD, PNG, JPEG, and its own KRA format.
	Python Scripting: Users can create custom plugins and improve Krita's functionality using the Python API
Adobe Photoshop	Generative Fill & Expand: Allows for the use of text prompts to assist with image editing.
	Generate Image: Inbuilt image generator for easy image creation.
	Remove Tool: Easy to use tool that removes unwanted objects.
PhotoPea	**Professional Editing Tools:** Includes brush tools, clone stamps, spot healing, and patch tools.
	Wide Format Support: Opens and saves in formats such as PSD, PNG, JPEG, GIF, PDF and SVG.
	AI-Powered Tools: Allows for simpler ease of use.

5. Identify a unique selling point for each image editing program. This might include its licensing or cost, its feature set, ease-of-use, or something else. (Write one or two sentences for each program)

Image Editing Software	Unique Selling Point
Krita	Free to use with a focus on digital painting, illustration and 2D animation. Krita use artist centric features and brushes for better results.
Adobe Photoshop	Requires an Adobe subscription to use because of its professional phot editing retouching and design capabilities. Photoshop is worth the price if needed for a more professional setting.
Photopea	Free web based phot editing, which is easily accessible. A great alternative to Adobe Photoshop due to its cost minimisation.

6. For each image editing program identified, list sources of information for users to refer to. This can be online technical information provided by the developer, how-to articles and blogs, YouTube channels, or textbooks.
   Identify at least 2 different sources for each engine.

Image Editing Software	Sources of Information
Krita	https://krita-artists.org/c/support/6
	https://krita-artists.org/
Adobe Photoshop	https://www.adobe.com/au/learn/photoshop
	https://www.youtube.com/watch?v=ZbvLJ5XtPpA
Photopea	https://blog.photopea.com/

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	https://www.youtube.com/playlist?list=PLkmh03xz-SPtYqEZSRHfwHriy_1VZ97wn

Audio Software

7. Identify 3 different programs used for editing audio.
   These may be PC applications, web services, or mobile apps.
   Research each program and provide a list of features (at least 3) for each.
   One of these programs must be Audacity.

Audio Editing Software	Features
Audacity	**Recording and Input:** Allows for recording and user input to be edited.
	Editing and Manipulation: Easy to use editing that allows for sound editing.
	Import/Export and Formats: Support high quality audio in WAV, MP3 and FLAC
GarageBand	**Multi-track Recording:** Allows for the recording of audio across multiple tracks.
	Sound Library: Access to multiple royalty free audio.
	Flex Time: Adjusts the timing of audio and instruments.
Adobe Audition	**Multitrack Editor:** Allows users to mix audio, apply effects and control audio projects.
	Waveform & Spectral Display: **Allows for precise editing of audio files.**
	Remix and Auto-Ducking: **Uses inbuilt AI to assist with the audio generation process.**

8. Identify a unique selling point for each audio editing program. This might include its licensing or cost, its feature set, ease-of-use, or something else. (Write one or two sentences for each program)

Audio Editing Software	Unique Selling Point
Audacity	Completely free and open source, very easy to learn for beginners.
GarageBand	User friendly music creation that is also free to use.
Adobe Audition	Industry standard quality, that is pay to use. It is worth the cost if you are working in industry.

9. For each audio editing program identified, list sources of information for users to refer to. This can be online technical information provided by the developer, how-to articles and

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blogs, YouTube channels, or textbooks.
Identify at least 2 different sources for each engine.

Audio Editing Software	Sources of Information
Audacity	https://support.audacityteam.org/
	https://www.youtube.com/playlist?list=PL6wGbZVVzENlzXKZJz3D1XmOZeL91GncR
GarageBand	https://support.apple.com/garageband
	https://guides.library.ucsc.edu/DS/Resources/IntroToGarageBand
Adobe Audition	https://www.adobe.com/au/learn/audition
	https://www.youtube.com/playlist?list=PL6wGbZVVzENk8Z7knuBOFWGz2aTHk2tex

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